Experiment No : 9

Title: Write a C++ program that illustrates run time polymorphism by using virtual functions.

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Program:

```
#include <iostream>
using namespace std;
// Base class
class Shape {
public:
// Virtual function
virtual void draw() {
cout << "Drawing a shape." << endl;
}
virtual ~Shape() {} // Virtual destructor
};
// Derived class 1
class Circle : public Shape {
public:
// Override the virtual function
void draw() override {
cout << "Drawing a circle." << endl; }};
// Derived class 2
class Rectangle : public Shape {
public:
// Override the virtual function
void draw() override {
cout << "Drawing a rectangle." << endl; }};
int main() {
// Create objects
Shape shape;
Circle circle;
Rectangle rectangle;
// Call the virtual function
shape.draw();
circle.draw();
rectangle.draw();
// Using base class pointers
Shape* shapePtr = &circle;
```

```
shapePtr->draw();
shapePtr = &rectangle;
shapePtr->draw();
return 0;
}
```

Output :

